

ROBIN BILODEAU

Ottawa, Ontario, Canada

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EDUCATION

University of Ottawa

Ottawa, Ontario, Canada

Bachelor of Science in Computer Science and Mathematics

Sep 2024 – Apr 2028

- **Relevant Courses:** Intro to Computing 1, Intro to Computing 2, Linear Algebra, Differential Equations and Laplace Transforms, Discrete Structures, Elementary Real Analysis, Introduction to probability, Data Structures and Algorithms

WORK EXPERIENCE

Scotiabank — Global Markets

Toronto, Ontario, Canada

Quantitative Analyst Intern

Sep 2025 – Dec 2025

- Refactored a legacy pricing process into a modular architecture supporting multiple rates derivatives, including swaps, swaptions, caps, and floors
- Optimized performance bottlenecks in core analytical functions, leading to substantial runtime improvements
- Conducted in-depth testing and investigation of pricing models to identify product-dependent discrepancies and assess model validity, and visualized discrepancy differences across products to communicate findings clearly

RESEARCH EXPERIENCE

Privacy-Preserving Zero-Knowledge Credential Protocol — Advised by Prof. Adams

Ottawa, Ontario, Canada

Undergraduate Researcher

May 2026 – Present

- Designed and implemented a zero-knowledge proof protocol over Pedersen commitments enabling privacy-preserving equality testing on hidden attributes without credential disclosure
- Constructed a multi-scenario threat model for the protocol, enumerating adversarial conditions across trust levels and derived mitigations including stateless handshake design, ephemeral signing key separation, and Curve25519 cofactor clamping

Data-Driven Assessment of Machine Learning Models for Climate-Sensitive Wildfire Susceptibility in Alberta — Advised by Prof. Bonakdari

Ottawa, Ontario, Canada

Undergraduate Researcher

Feb 2026 – Present

- Implemented and compared 13 variants of the Group Method of Data Handling (GMDH) self-organizing algorithm for interpretable wildfire susceptibility detection
- Designed novel fractional-polynomial, basis-function, and look-back GMDH architectures to achieve superior balance between predictive accuracy and interpretability

Uncertainty-Aware Statistical Extensions for HDBSCAN Clustering — Advised by Prof. Smith

Ottawa, Ontario, Canada

Undergraduate Researcher

Apr 2025 – Oct 2025

- Developed a Monte Carlo sampling framework to repeatedly adjust probability rank matrices, yielding statistically stable core-distance estimates for clustering high-noise data, reducing sensitivity to outliers and improving cluster detection reliability
- Designed a fast Fourier-based deconvolution estimator to recover underlying data distributions from noise-contaminated observations, enhancing core distance estimates for robust HDBSCAN clustering

LEADERSHIP AND ACTIVITIES

SSC Case Study Competition - Parkinson's Disease Classification

Hamilton, Ontario, Canada

Team Member

Apr 2026 – Jun 2026

- Performed end-to-end data cleaning on a high-dimensional clinical dataset (500+ variables) from the Canadian Open Parkinson Network (C-OPN), including leakage analysis to identify and remove variables that directly encode diagnostic criteria for PD vs. atypical parkinsonism, and outcome-stratified MICE imputation to handle missing data
- Developed a random forest classifier achieving greater than 95% accuracy in distinguishing Parkinson's disease from atypical parkinsonism syndromes, using stratified k-fold cross-validation with fold splitting prior to imputation

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Fluent), French (Native)
- **Programming:** Python, R, Java, Typst, C++ (Beginner)
- **Tools:** Microsoft Word, Git, Visual Studio Code, Canva, Jupyter
- **Libraries:** matplotlib, numpy, Scikit-learn, pandas, yfinance
- **Athletics:** Boxing, Hockey, Soccer, Volleyball
- **Activities:** Tutoring, Mathematics Club, Undergraduate Research Club